EARTHQUAKES

GOO

1. Introduction to Earthquakes 🌍😳

Definitions:

- **Earthquake:** Defined as a tremor below the surface of the earth which causes shaking of the crust.
- **Seismic Waves:** A series of shocks that result from sudden earth movement (e.g., along a fault) spreading out from the focus.
- **Seismic Focus (Hypocentre):** The point of origin of the seismic waves in the interior of the earth. It originates deep at the fault line.
- **Epicentre:** The point on the earth's surface directly above the seismic focus.

Key Points:

- These devastating tremors are caused strictly by the stresses that break the earth's crust.
- About 20,000 earthquakes occur every year across the world, according to scientists.
- They are caused by the strain in the earth's crust; when the crust is unable to accommodate this strain, it results in a sudden release of tremendous energy in the form of a violent shock.
- Earthquakes spread out from the focus and cause widespread destruction, heavily concentrated in areas along the edges of lithospheric plates (e.g., down the western edge of the Pacific Ocean).
- Most earthquakes have their seismic focus between 8 to 35 km in depth.

2. Causes of Earthquakes 🌋💥

Key Points (The 4 Main Factors):

- **(i) Volcanoes:** Volcanic eruptions and earthquakes are heavily interrelated; hot gases push violently upward against the crust. An earthquake follows a volcanic eruption, and vice versa.
    - *Critical Example:* The violent eruption of Krakatoa between Sumatra and Java caused a severe earthquake and generated a Tsunami that killed about 30,000 people in the vast region.
- **(ii) Plate Tectonics:** When movable lithospheric plates slip past or collide against each other, their edges produce faults along the lines of weakness. These tectonic earthquakes are the most frequent and immensely powerful.
    - *Critical Example:* In 2001, a severe earthquake hit Bhuj in Gujarat due to the lowering of the Indian plate below the Asiatic plate.
- **(iii) Folding and Faulting:** Compressional and tensional forces cause internal horizontal and vertical movements, displacing rocks.
    - *Critical Example:* The 1906 San Andreas Fault earthquake in California (magnitude 7.8) slipped along a 430 km segment, causing a devastating fire and a destruction zone 400 miles long and 25 miles wide.
- **(iv) Man-made Causes:** Human activity directly triggers earthquakes. This includes the construction of huge dams, mining, rock blasting, and **nuclear tests / nuclear explosions** near fault zones.
    - *Critical Example 1:* The Koyna earthquake in Maharashtra was caused by the construction of the Koyna dam in a seismically active region.
    - *Critical Example 2:* Chemical wastes dumped into the ground and underground explosions caused several earthquakes in Colorado (USA) in the 1960s.

3. Anatomy of an Earthquake (Seismic Waves) 〰️😳

Key Points (Integrating Handwritten Wave Physics!):

- Tension builds up until it is released as powerful vibrations traveling in three distinct types of waves.

- **Body Waves:** Both P-waves and S-waves are classified as deep interior body waves.
- **Refraction & Shadow Zones:** When seismic waves travel between two different mediums inside the earth, they undergo **refraction**. This bending creates specific "shadow zones" where absolutely **no P-waves and no S-waves** are detected.

Table of Differences (The 3 Seismic Waves) 🆚

Wave Type	Characteristics & Movement 🏃	Mediums it can Travel Through 🧱💧
1. P-Waves (Primary / Push)	First to be recorded. They are compressional and **longitudinal** (particles vibrate strictly in the direction of wave movement, like sound waves).	Can pass through ALL mediums: solids, liquids, and gases.
2. S-Waves (Secondary / Shear)	Recorded after P-waves. They are distortional and **transverse** (particles vibrate at right angles to wave movement). They are highly destructive.	Travel through the interior but **cannot be transmitted by liquids**.
3. L-Waves (Surface / Long)	Derived directly **from body waves** hitting the surface. They are the last to arrive and travel along the crust causing a **sideways, rolling** motion. (Includes Love and Rayleigh waves).	Do not pass through liquids. They make inhabitants feel the ground motion and are the **most destructive**.

4. Measurements of Earthquakes 📏📈

Definitions:

- **Seismograph:** An instrument that records the direction of movement and passage of waves at a particular point.
- **Richter Scale:** A mathematically precise scale (1 to 9) that measures the absolute intensity and amount of energy released during an earthquake.
- **Mercalli Scale:** A 12-point scale that grades earthquakes strictly on the basis of observed physical effects.

Key Points (Exhaustive Detail):

- **How a Seismograph Works:** It utilizes a heavy mass that relies on **inertia** to resist motion. As the earth moves during a quake, a pen attached to it vibrates, recording movements on a travelling strip of paper. It calculates the difference in arrival times of P and S waves to determine the exact distance to the epicentre.
- **Richter Scale Math:** The Richter scale is the most commonly used. Each single point upward represents a **10-fold increase** in the amplitude of shock waves (e.g., an earthquake of magnitude 6 is exactly 10 times stronger than magnitude 5).

Things to Remember (Major Earthquake Data) 🧠

- **Assam, India (1897):** Magnitude 8.7.
- **Gujarat, India (2001):** Magnitude 7.7 (caused the greatest damage).
- **Nepal (April 25, 2015):** Magnitude 7.8. Killed >8,000 people, flattened entire villages, and destroyed UNESCO World Heritage sites in the Kathmandu Valley. Aftershocks continued for 15-20 minutes.
- **Nepal Aftershock (May 12, 2015):** A major aftershock of 7.3 magnitude killing >200 and injuring 2500.

The "Why?" Section ❓

- **Why are earthquakes considered both constructive and destructive?** In geological terminology, earthquakes are part of endogenic forces and are considered *constructive* movements of the earth's crust (building new

relief). However, from a human point of view, they are completely *destructive* because they cause immense loss to property and life. *(Note: The Page-Shift Protocol successfully captured this final paragraph before the page cut off!)* 🧩

5. Effects of Earthquakes (Constructive Effects) 🏗️🌍

Key Points:

- **1. Energy release:** Earthquakes help the earth release its stored-up energy. The majority of earthquakes occur around the plate margins. This energy release critically helps to keep the earth in good shape. ⚡
- **2. Landforms:** On account of both vertical and lateral displacement of the earth's crust, earthquakes may raise or lower parts of the earth, especially near the seashore.
    - *Critical Example:* Sagami Bay of Japan was uplifted 200 m in 1923. 📈
    - Landslides triggered by earthquakes cause the formation of new lakes, as has happened at many places in the Himalayan region. 🏞️

6. Effects of Earthquakes (Destructive Effects) 💥🏚️

(Note: Page-Shift Protocol successfully engaged to seamlessly stitch the Tsunami and Flash Floods data across pages 76 and 77!) 🧩📖

Definitions:

- **Tsunami:** A large destructive ocean wave generated by any disturbance that rapidly displaces a large mass of water (earthquake, volcanic eruption, landslide, or meteorite impact). The name is from Japanese words: 'Tsu' (harbour) and 'nami' (wave), meaning harbour waves. 🌊

Key Points (Exhaustive Detail):

- **1. Collapse of structures:** Human beings have settlements in active earthquake zones. These houses and structures often collapse causing great loss of property (e.g., the destruction caused by a tsunami). 🏚️

- **2. Submergence:** The forces of uplift and subsidence cause submergence of coastal parts.
    - *Critical Example:* Dwarka in Gujarat now lies completely submerged under the sea. 🌊
- **3. Course of Rivers:** Earthquakes have changed the course of rivers in the past, rendering many areas entirely unsuitable for irrigation and agriculture. رود
- **4. Danger to human life:** Severe earthquakes have killed hundreds to lakhs of people. They die on account of house collapse, falling trees, poles, bridges, or by landslides and floods caused by earthquakes. ☠️
- **5. Landslides:** Shock waves destabilise loose rocks leading to landslides. This completely damages buildings, disturbs the transport system, and is common in the Himalayas, Alps, and Andes. 🪨
- **6. Fires:** Fires break out due to inflammable material thrown out, broken gas pipes, and the bursting of gas cylinders. Damaged oil refineries and live electric wires catch fire, heavily damaging life and property in townships and factory complexes. 🔥
- **7. Tsunami (The Deadliest Effect):** The most common cause is an undersea earthquake. They occur most frequently in the Pacific Ocean but are possible wherever large bodies of water are found (including inland lakes where they can be caused by landslides).
    - *Critical Example:* The 2004 Indian Ocean earthquake triggered a series of lethal tsunamis on December 26, 2004, that killed approximately 2,75,000 people (more than 168,000 in Indonesia alone), making it the deadliest tsunami in recorded history. 📌
- **8. Flash Floods:** Strong shock waves strictly damage dams and embankments. The water stored there spreads speedily, causing massive flash floods. Furthermore, debris produced blocks the flow of rivers, diverting courses and causing massive damage. 🌊🚧

The "Why?" Section ❓

- **Why did a Tsunami strike Java and Sumatra in 2018 if there wasn't a massive tectonic plate earthquake?** On December 22, 2018, a tsunami (>300 deaths) was caused strictly by *undersea landslides* due to the ongoing volcanic eruption of the Anak Krakatau volcano! Indonesia is an archipelago on the Pacific Ring of Fire formed by molten magma pressure, making it highly susceptible to these chain-reaction events. 🌋🌊

7. Distribution of Earthquakes 🗺️😳

Key Points:

- Most earthquakes originate from plate boundaries. In regions where ocean and land plates converge, submarine ridges are formed. The underlying rocks are unstable, and movement triggers earthquakes in continental areas.
- **Belt 1: The Circum Pacific Mountain Belt**
    - This belt has **70%** of all earthquakes. 📈
    - A part of the San Andreas Fault in the USA lies in this belt, making California highly prone to earthquakes. us
    - Japan, the Philippines, and Indonesia all form parts of this belt and are prone to severe earthquakes. 📌
- **Belt 2: The Mid-world Mountain Belt**
    - Stretches from Eastern Europe covering Alpine-Himalayan ranges in Europe and Asia.
    - About **20%** of all earthquakes occur in this zone. 📈
- **Belt 3: The Mid-Atlantic Ridge**
    - Comprises areas along mid-oceanic ridges and islands near the ridges of the Atlantic Ocean.
    - The remaining **10%** of earthquakes occur here. Two tectonic plates move in opposite directions, causing moderate earthquakes.

- The Rift Valley of East Africa is considered an extension of this belt. 📌

Table of Differences (Earthquake Measurement Scales) 🆚

Feature	Richter Scale 📏	Moment Magnitude (Mw) Scale 📐
History	Standardised magnitude measurements in the 1930s.	Developed later based on the exact same principle as Richter.
Accuracy	Good for relative sizes, but less accurate for massive events.	Considered the **most accurate and reliable method** for measuring large earthquakes.
What it Measures	Amplitude/power of the earthquake wave.	Measures the **total amount of energy released** during an earthquake.

8. Predicting Earthquakes 🔮📈

Key Points:

- In the past, prediction was left to astrologers and mystics. Today it is a respectable scientific pursuit. 🔬
- The Russians were the first to discover P and S seismic waves in the 1960s on the basis of which predictions were made. These waves are still the only means available to seismologists to predict earthquakes. 〰️
- The theory of Plate Tectonics offers another means of prediction on scientific lines. 🗺️
- The Japanese use methods of measuring changes in sea level and variations in Earth's magnetic field to predict earthquakes. 🌊🧲

- **CRITICAL LIMITATION:** However, no method is still foolproof, as has been proved by recent devastating earthquakes in Japan, India, and the USA. 🚫

Things to Remember (Top 5 Earthquakes Ever Recorded by M_w) 🧠

Rank	Location	Date	Magnitude (Mw)
1	Valdivia, Chile	22-05-1960	9.5
2	Alaska, USA	27-03-1964	9.2
3	Sumatra	26-12-2004	9.1 to 9.3
4	Tohoku (Japan)	11-03-2011	9.1
5	Kamchatka, Russia	05-11-1952	9.0